



Navigating the gambit: Unfavorable market responses to AI-based service patents in manufacturing firms

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ABSTRACT

Drawing on transaction cost economics and organizational information processing theory, we challenge the espoused benefits of AI in servitization by demonstrating that stock market reactions to service technology patents with high AI scores are negative. In a sample of 1306 service technology patents and 758,143 non-service technology patents from 3899 manufacturing firms (1980–2020), we find that idiosyncratic volatility mitigates this negative reaction, while tangibility shows no such effect. Decade-wise analysis reveals effects are significant only in the 2010s and 2020s, suggesting evolving market perceptions of AI-driven servitization. Results hold across matched-pair sampling, non-linear effects testing, fixed-effects individual slopes, and alternative moderator analyses. This study cautions against assuming short-term market value of AI service innovations in manufacturing.

1. Introduction

The manufacturing industry is undergoing a servitization revolution, shifting focus from solely producing goods to offering a comprehensive portfolio of services alongside physical products (Baines et al., 2009; Kohtamäki et al., 2019a; Rabetino et al., 2018; Raddats et al., 2019; Vandermerwe and Rada, 1988). Artificial Intelligence (AI)-powered service technology empowers manufacturing firms to not only enhance efficiency and productivity but also embark on a path to deliver exceptional customer experiences and establish long-term value-creation strategies (Aker et al., 2023; Bustinza et al., 2018; Gebauer et al., 2021; Sjödin et al., 2023). AI service technology can help manufacturers automate repetitive tasks, streamline workflows, and optimize resource utilization, leading to tangible improvements in operational efficiency, cost reduction, and increased productivity (Kharlamov and Parry, 2021; Paschou et al., 2020). AI-powered services can handle logistics, automate maintenance scheduling, and personalize customer interactions, ultimately contributing to a seamless and efficient service delivery model. By responding quickly to inquiries and resolving issues efficiently, AI could enhance customer satisfaction and facilitate a positive experience throughout the service lifecycle.

Due to challenges in the implementation of AI in services, in general, and servitization in particular, we ask whether AI service patents lead to negative stock market reaction and whether this negative reaction can be mitigated by factors that improve organizational information

processing—idiosyncratic volatility and tangibility. From an organizational information processing theory (OIPT) perspective (Galbraith, 1973; Kroh et al., 2018), the increased complexity of servitization introduced by OIPT posits that organizations have a finite capacity to process and utilize information, and excessive complexity can overload this capacity, leading to decision challenges, recombination of operational resources, decreased efficiency, and lower performance. In the context of servitization, AI-based service patents introduce an added layer of complexity that can divert resources away from core servitization activities, such as developing and delivering value-adding services and building customer relationships (Jorzik et al., 2024; Stock and Tatikonda, 2008). AI-based service patents can be attributed to the additional cognitive and logistical demands placed on manufacturing firms (Wan et al., 2020). Additionally, managing a portfolio of AI-powered service patents requires specialized expertise in managing manufacturing and service-related resources. Moreover, the ambiguity and evolving nature of AI technology can further complicate the interpretation and rebundling of operational resources.

From a transaction cost economics and organizational information processing theory (OIPT) perspective, idiosyncratic volatility (Dai, 2021; Xu and Malkiel, 2003) and tangibility (Kamp and Parry, 2017; Luiz Corrêa et al., 2007) can help mitigate the negative reactions associated with the approval of AI-based service patents. Idiosyncratic volatility, also or unsystematic risk, can help manufacturing firms maintain a focus on their core servitization activities thus reducing

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